Credit Risk Analysis & Predictive Modeling

End-to-End Project Documentation

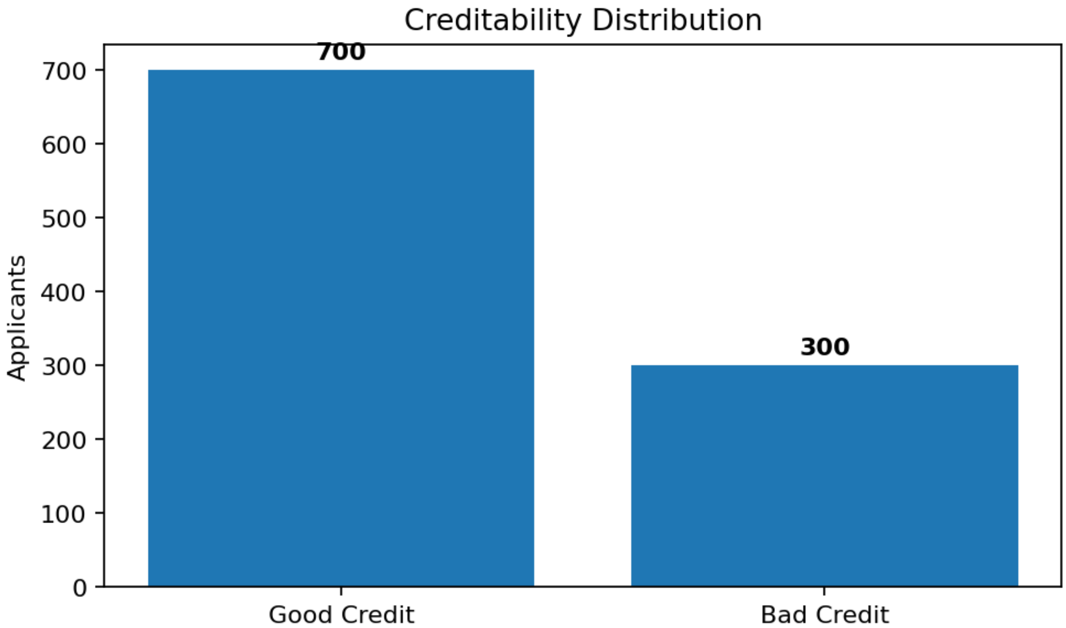
1,000 historical credit applications | Python, Pandas, Scikit-learn, SciPy, Matplotlib, Seaborn

1. Project Objective

Analyze historical credit applications to identify characteristics associated with creditability and evaluate machine-learning models that distinguish good-credit from bad-credit applicants. The workflow combines EDA, statistical testing, predictive modeling, cross-validation, threshold analysis, feature importance, and business interpretation.

2. Dataset Overview

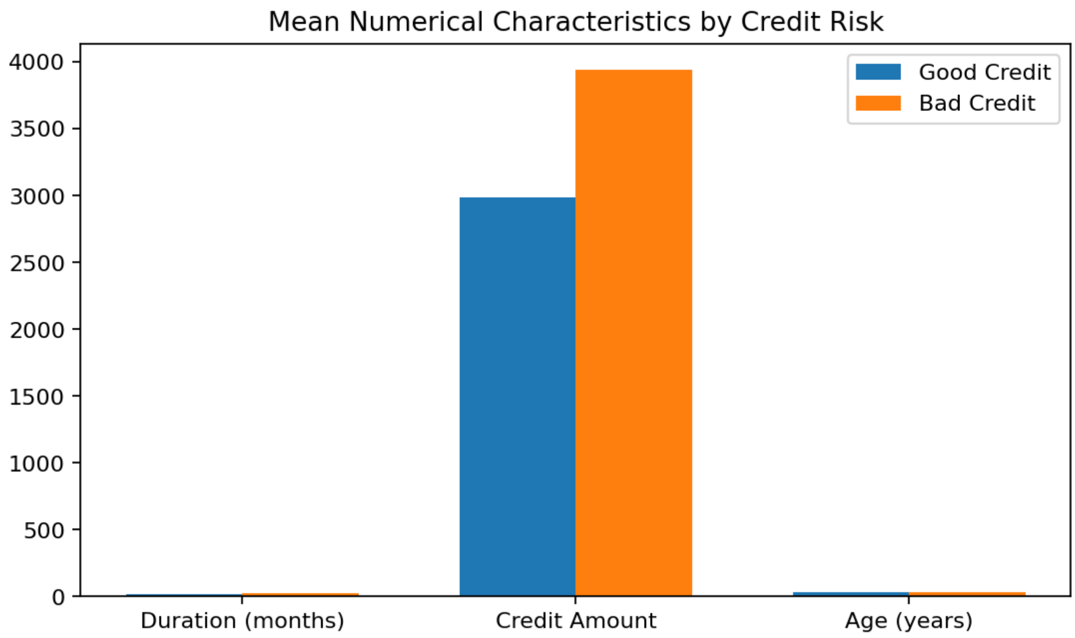
- 1,000 records and 21 columns.
- 700 good-credit applicants (70%) and 300 bad-credit applicants (30%).
- 0 missing values and 0 duplicate rows.
- 20 original predictors; many integer-coded fields represent categories.



3. Data Quality & Preparation

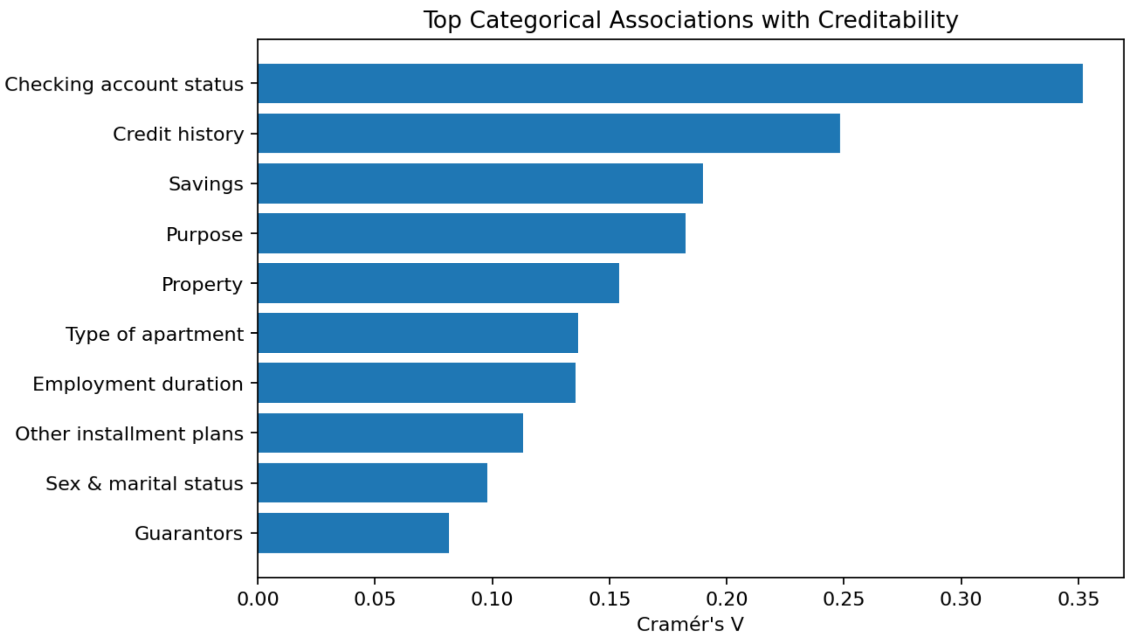
EDA helper columns were excluded from modeling by explicitly selecting the 20 original predictors. Duration, Credit Amount, and Age were treated as numerical variables; the remaining 17 predictors were treated as categorical codes. Numerical variables were standardized and categorical variables one-hot encoded inside a Scikit-learn Pipeline and ColumnTransformer.

4. Exploratory Data Analysis



- Bad-credit mean duration: 24.86 vs 19.21 months.
- Bad-credit mean credit amount: 3,938.13 vs 2,985.44.
- Bad-credit mean age: 33.96 vs 36.22 years.
- These are observational differences and do not establish causality.

5. Statistical Association Testing



Checking account status had the strongest categorical association (Cramér's V=0.3517), followed by credit history (0.2484), savings (0.1900), and purpose (0.1826).

Feature	V	p-value
Checking account status	0.3517	<0.0001
Credit history	0.2484	<0.0001
Savings	0.1900	<0.0001
Purpose	0.1826	0.0001
Property	0.1540	<0.0001
Type of apartment	0.1367	0.0001

Employment duration	0.1355	0.0010
Other installment plans	0.1133	0.0016
Sex & marital status	0.0980	0.0222
Guarantors	0.0815	0.0361
Foreign worker	0.0763	0.0158
Installment percent	0.0740	0.1400
No. of credits at bank	0.0517	0.4451
Occupation	0.0434	0.5966
Telephone	0.0342	0.2789
Present residence	0.0274	0.8616
No. of dependents	0.0000	1.0000

6. Machine-Learning Setup

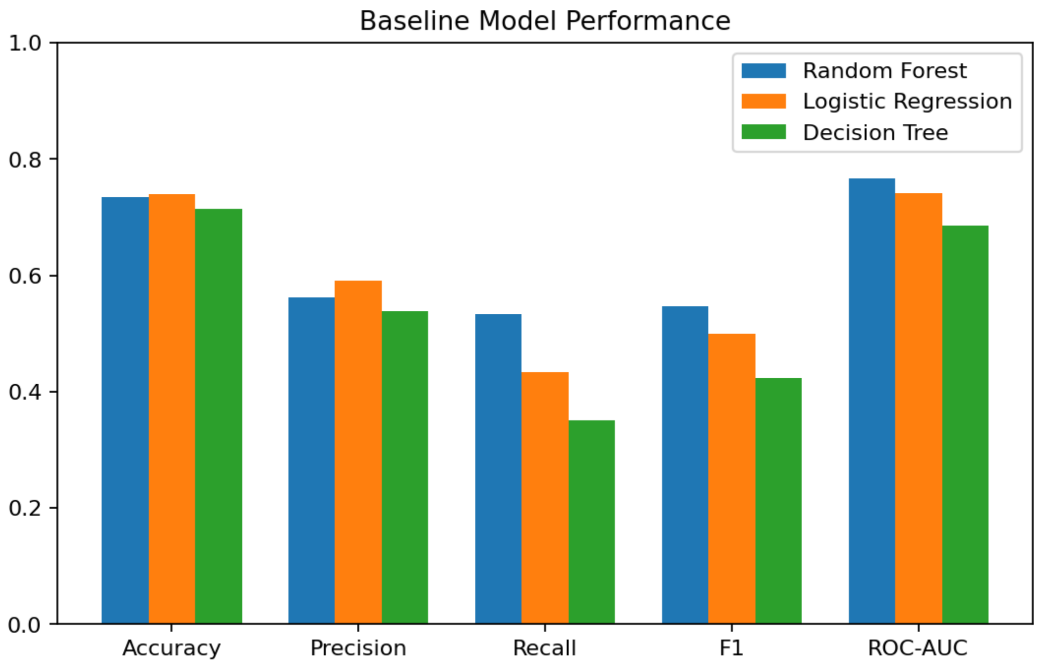
• Target: 1 = Bad Credit, 0 = Good Credit.

• 80/20 stratified train-test split, random_state=42.

• Models: Logistic Regression, Decision Tree, Random Forest.

• Metrics: Accuracy, Precision, Recall, F1 Score, ROC-AUC.

7. Baseline Model Evaluation



Model	Acc	Prec	Recall	F1	AUC
Random Forest	0.735	0.561	0.533	0.547	0.767
Logistic Regression	0.740	0.591	0.433	0.500	0.742
Decision Tree	0.715	0.538	0.350	0.424	0.685

Random Forest was the preferred baseline because it achieved the highest recall, F1-score, and ROC-AUC, although Logistic Regression had slightly higher accuracy and precision.

8. Cross-Validation

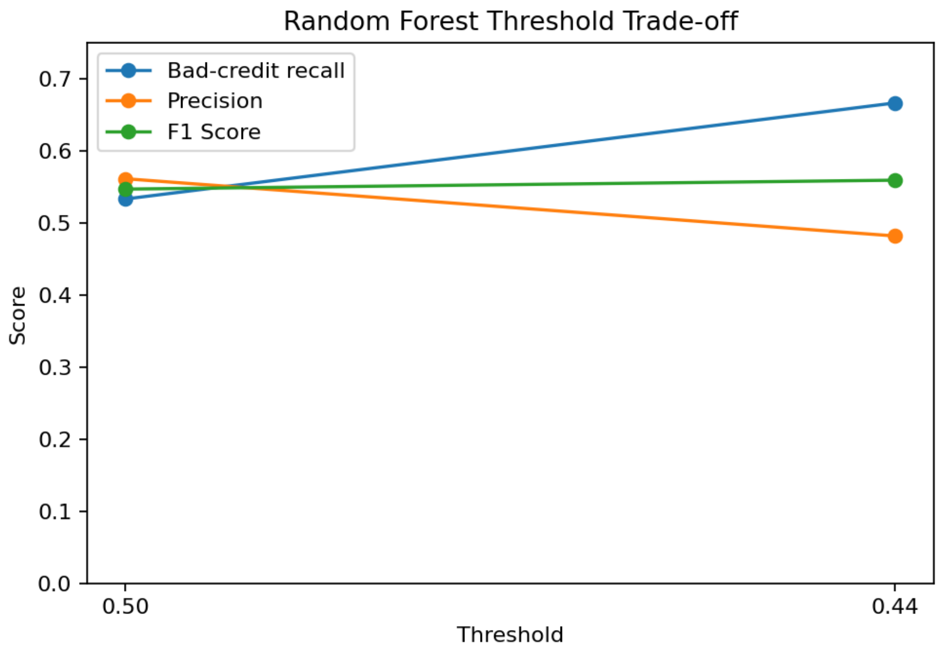
Metric	Mean	Std Dev
Accuracy	0.7480	0.0319
Precision	0.5849	0.0586
Recall	0.5633	0.0748
F1 Score	0.5717	0.0567
ROC-AUC	0.7852	0.0447

Five-fold stratified cross-validation for baseline Random Forest produced mean ROC-AUC 0.7852 ± 0.0447.

9. Hyperparameter Tuning

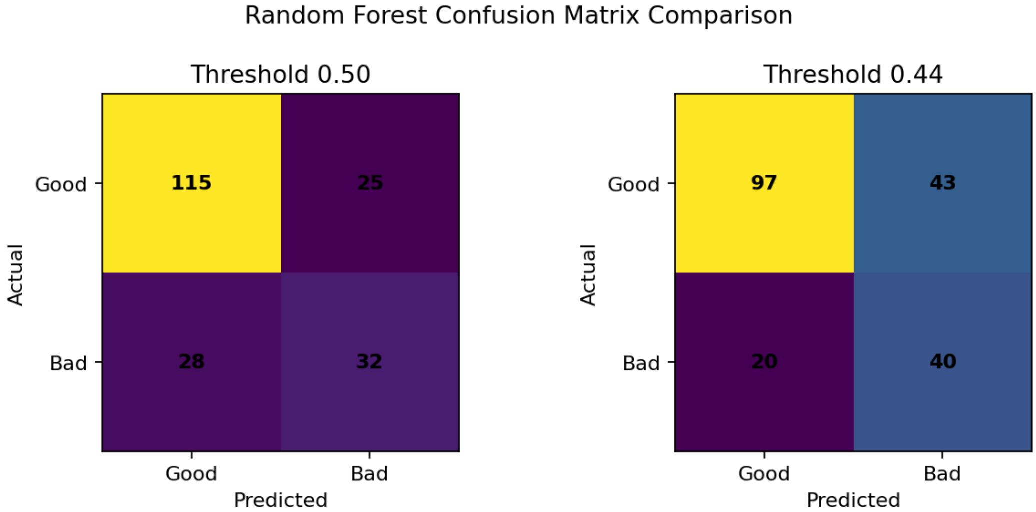
RandomizedSearchCV evaluated 40 configurations across five folds. Best CV ROC-AUC was 0.7958 with n_estimators=300, min_samples_split=15, min_samples_leaf=4, max_features=log2, max_depth=None, and balanced_subsample class weighting. On the untouched test set, tuned ROC-AUC was 0.7542 versus 0.7670 for baseline, so the baseline was retained.

10. Threshold Analysis



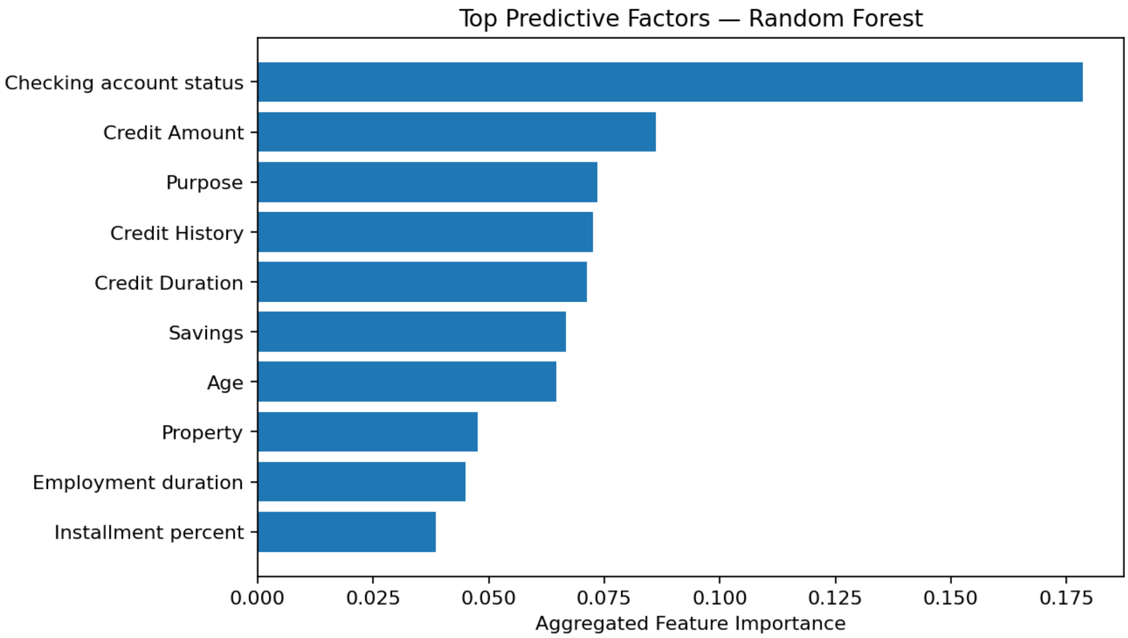
A CV-selected threshold of 0.44 increased bad-credit recall from 53.3% to 66.7% on the held-out test set, while precision decreased from 56.1% to 48.2% and accuracy from 73.5% to 68.5%.

11. Confusion Matrix



Threshold 0.50: [[115, 25], [28, 32]]. Threshold 0.44: [[97, 43], [20, 40]]. The lower threshold catches 8 additional bad-credit applicants in the 200-row test set, while producing 18 additional false positives.

12. Feature Importance



Feature	Importance
Checking account status	0.178504
Credit Amount	0.086125
Purpose	0.073441
Credit History	0.072626
Credit Duration	0.071242
Savings	0.066684
Age	0.064685
Property	0.047533
Employment duration	0.044982
Installment percent	0.038465

13. Integrated Findings

• Checking account status was strongest in both statistical association and Random Forest importance.

• Credit history, savings, purpose, property, credit amount, and credit duration were meaningful signals.

• Bad-credit applicants had higher average credit amounts and longer average durations.

• Occupation and number of dependents were weak in categorical association analysis.

• Agreement between EDA, statistical testing, and model importance strengthened the analytical conclusions.

14. Business Recommendations

• Give appropriate analytical attention to checking-account status and credit history.

• Consider credit amount and repayment duration together.

• Use statistically informative categorical variables as supporting signals rather than isolated decision rules.

• If catching more potentially risky applicants is the priority, a lower threshold can be considered with additional review.

• Track precision, recall, F1, and ROC-AUC alongside accuracy.

15. Limitations

• Only 1,000 historical applications; generalization is uncertain.

• Encoded categorical variables require careful interpretation.

• Feature importance does not establish causality.

• Test ROC-AUC of 0.767 indicates useful but imperfect discrimination.

• Threshold selection did not include real financial cost estimates.

• Production use would require representative current data, calibration, monitoring, fairness assessment, governance, and further validation.

16. Key Lessons Learned

• Categorical integer codes are not automatically continuous variables.

• Compare within-group rates, not only raw counts.

• Use statistical significance and effect size together.

• Cross-validation helps assess stability.

• Hyperparameter tuning does not guarantee better held-out performance.

• Classification thresholds should reflect business objectives.

• Translate model outputs into business-level insights.

17. Final Summary

The project demonstrates an end-to-end analytics workflow from data-quality checks and EDA through statistical testing, machine learning, cross-validation, tuning, threshold analysis, feature importance, and business recommendations. The preferred overall baseline is Random Forest with held-out test ROC-AUC 0.767. A 0.44 threshold provides a risk-sensitive operating point with 66.7% bad-credit recall while making the precision and accuracy trade-offs explicit.